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UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-29 & MVD-30 20kV Centralized Panel Room WICA SCHEMATIC DIAGRAM SBC-W INCOMING H01,H08															
				Date Prep. 15.11.2018		PROJECT TITLE				COVER SHEET		Scale		NTS	
				Prepared YR		MV Switchgear for MNA Serang Plant				MVD-29 & MVD-30 20kV Centralized Panel Room WICA		Customer		PT Multimas Nabati Asahan	
00		Issued for Approval		16.01.19		YR		Checked AW		PT. ABB Sakti Industri		Drawing Nbr.		18.042-MNA-304.10	
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	Index of Sheet MVD-29 & MVD-30 20kV Centralized Panel Room WICA SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01,H08
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 304
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(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)

-BGB5	POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE ENERGIZED
-BGB6	POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE EXCLUDED MECHANICALLY
-BGB7	POSITION SWITCHES OF CIRCUIT-BREAKER ON SWITCHGEAR
-BGB8...-BGB10	POSITION SWITCHES OF CIRCUIT-BREAKER
-BGB11	POSITION SWITCH OF CIRCUIT-BREAKER OPERATING FOR MANUAL OPENING
-BGB12	PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN OPEN POSITION
-BGB13	PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN CLOSED POSITION
-BGD	POSITION SWITCH OF L.V. INSTRUMENT COMPARTMENT DOOR.
-BGD1	POSITION SWITCH OF CIRCUIT-BREAKER COMPARTMENT DOOR
-BGD2	POSITION SWITCH OF CABLE COMPARTMENT DOOR
-BGE1	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION
-BGE2	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN CLOSED POSITION
-BGE3	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)
-BGE4	POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE ACTUATED BY ELECTRIC MOTOR
-BGE5	POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 ACTUATED BY ELECTRIC MOTOR
-BGE7	POSITION SWITCH SIGNALLING EARTHING SWITCH NOT IN MANUAL OPERATION
-BGE8	POSITION SWITCHES OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH- DISCONNECTOR -QBS
-BGE11	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION
-BGE12	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN CLOSED POSITION
-BGE13	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)
-BGE14, -BGE15	POSITION SWITCHES FOR OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS, ACTUATED BY ELECTRIC MOTOR
-BGF	POSITION SWITCHES OF MEDIUM VOLTAGE FUSES
-BGI1	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION
-BGI2	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN CLOSED POSITION
-BGI3	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)
-BGI4	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR
-BGI5	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR
-BGI6	POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN EMERGENCY MANUAL OPERATION
-BGI7	POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN MANUAL OPERATION
-BGI8	POSITION SWITCHES OF SWITCH-DISCONNECTOR -QBS
-BGI9	POSITION SWITCHES OF OPERATION OF SWITCH-DISCONNECTOR -QBS ACTUATED BY ELECTRIC MOTOR
-BGI11	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION
-BGI12	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN CLOSED POSITION
-BGI13	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)
-BGI14, -BGI15	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR
-BGI21	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION
-BGI22	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN CLOSED POSITION
-BGI23	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)
-BGI24, -BGI25	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR

ABB EPMV

PT. ABB Sakti Industri

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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS							A
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR							
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.							
	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT							
B	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							B
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT							
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT							
	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT							
C	-BGK	POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET							C
	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST							
	-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.							
	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
D	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							D
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1							
	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2							
	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3							
E	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							E
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1							
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2							
	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3							
F	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							F
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1							
	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2							
G	-BM	HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							G
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES							
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS							
	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR							
H	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							H
	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK							
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							
I	-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							I
	-BT	THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS							
	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS							
	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY							
J	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS							J
	-BU	BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS							
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS							
	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)							
K	-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS							K
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES							
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS							
L								-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							L
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							
M								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							M
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS							
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS							
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS							
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS							
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK							
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING							
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH							
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)							
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)							
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES							
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR							
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES							
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS							
	-PFF	FLAG RELAYS						-RAR	RESISTORS							
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS							
C	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING							
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT							
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE							
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION							
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD							
	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1							
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2							
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2							
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS							
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4							
D	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5							
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES							
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS							
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS							
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS							
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS							
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES							
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS							
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS							
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES							
E	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS							
	-PGV	VOLTMETERS						-TB	CONVERTER							
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE							
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE							
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE							
	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING							
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT							
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE							
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER							
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)														
F	-QBD4	TWO-WAY DISCONNECTOR														
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER														
					Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-29 & MVD-30 20kV Centralized Panel Room WICA		Scale	NTS		= H01,H08
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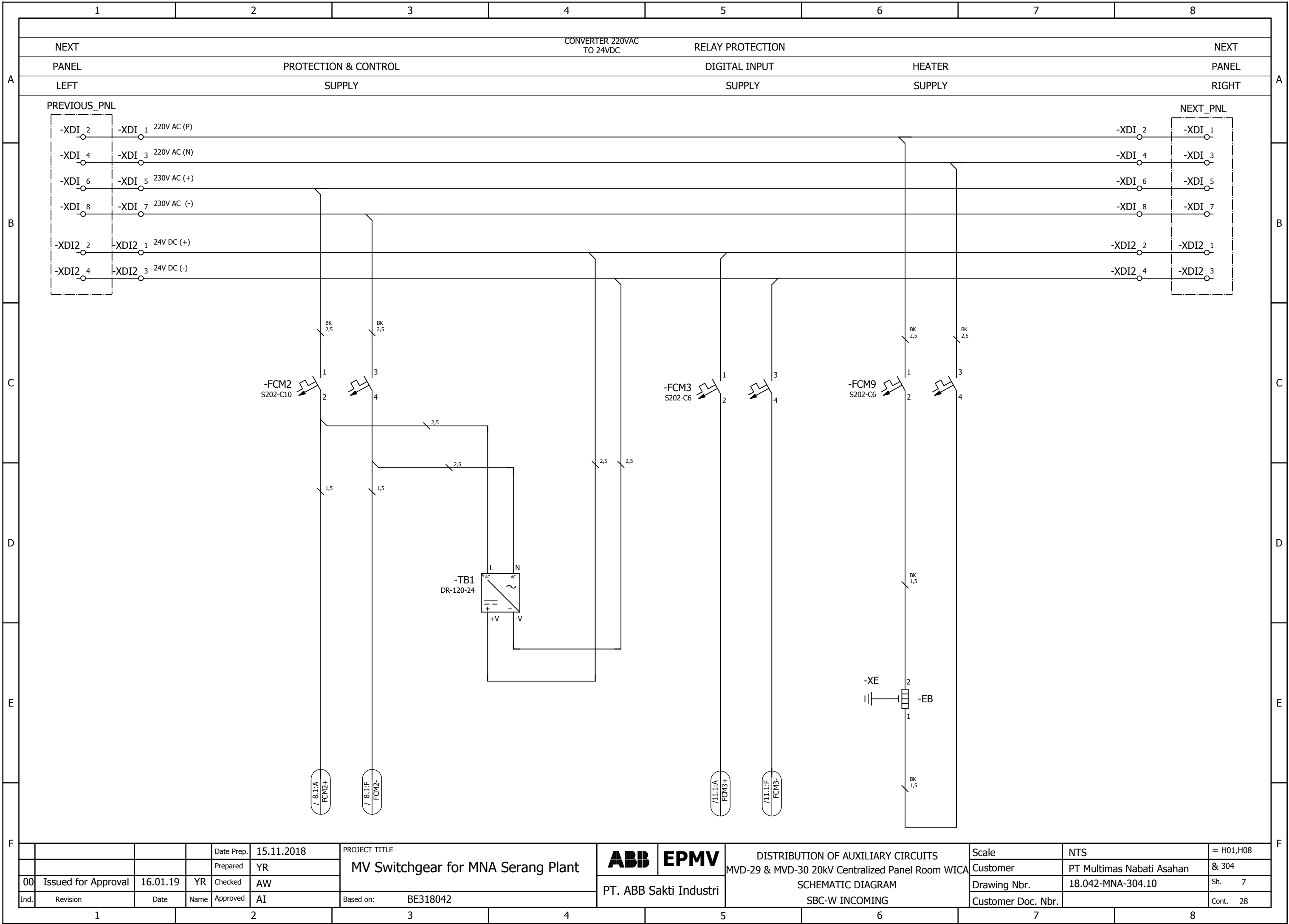
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A	-TFA	ACTIVE POWER TRANSDUCERS	-XDI1	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS
	-TFC	CURRENT TRANSDUCERS	-XDI2	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING
	-TFF	FREQUENCY TRANSDUCERS	-XDI3	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION
	-TFJ	ACTIVE ENERGY TRANSDUCERS	-XDI4	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS
	-TFK	REACTIVE ENERGY TRANSDUCERS	-XDI6	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS
	-TFM	MULTIFUNCTION TRANSDUCERS		
	-TFP	POWER-FACTOR TRANSDUCERS		
	-TFQ	REACTIVE POWER TRANSDUCERS		
	-TFS	SIGNAL CONVERTERS		
	-TFV	VOLTAGE TRANSDUCERS		
B	-WA	BUSBARS	-XDI7	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS
	-WBC	POWER CABLES	-XDI8	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS
	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.	-XDI9	TERMINAL BLOCK FOR SPECIAL USE
	-WGC	CONTROL CABLES	-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)	-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS
	-WH	OPTICAL FIBRES	-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS	-XDS	CURRENT SOCKETS
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1	-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2	-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3	-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK
C	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER	-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS	-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1
			-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2
			-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3
			-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)	-XDX	SUPPORT TERMINAL BLOCK
			-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER	-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS	-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS	-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE
D	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR	-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET		
	-XDC	CUSTOMER TERMINAL BLOCK		
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER		
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR		
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT		
E	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS		
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER		
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE		
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE		
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2		
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2		
	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS		
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR		
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS		
F	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS		
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)		
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)		

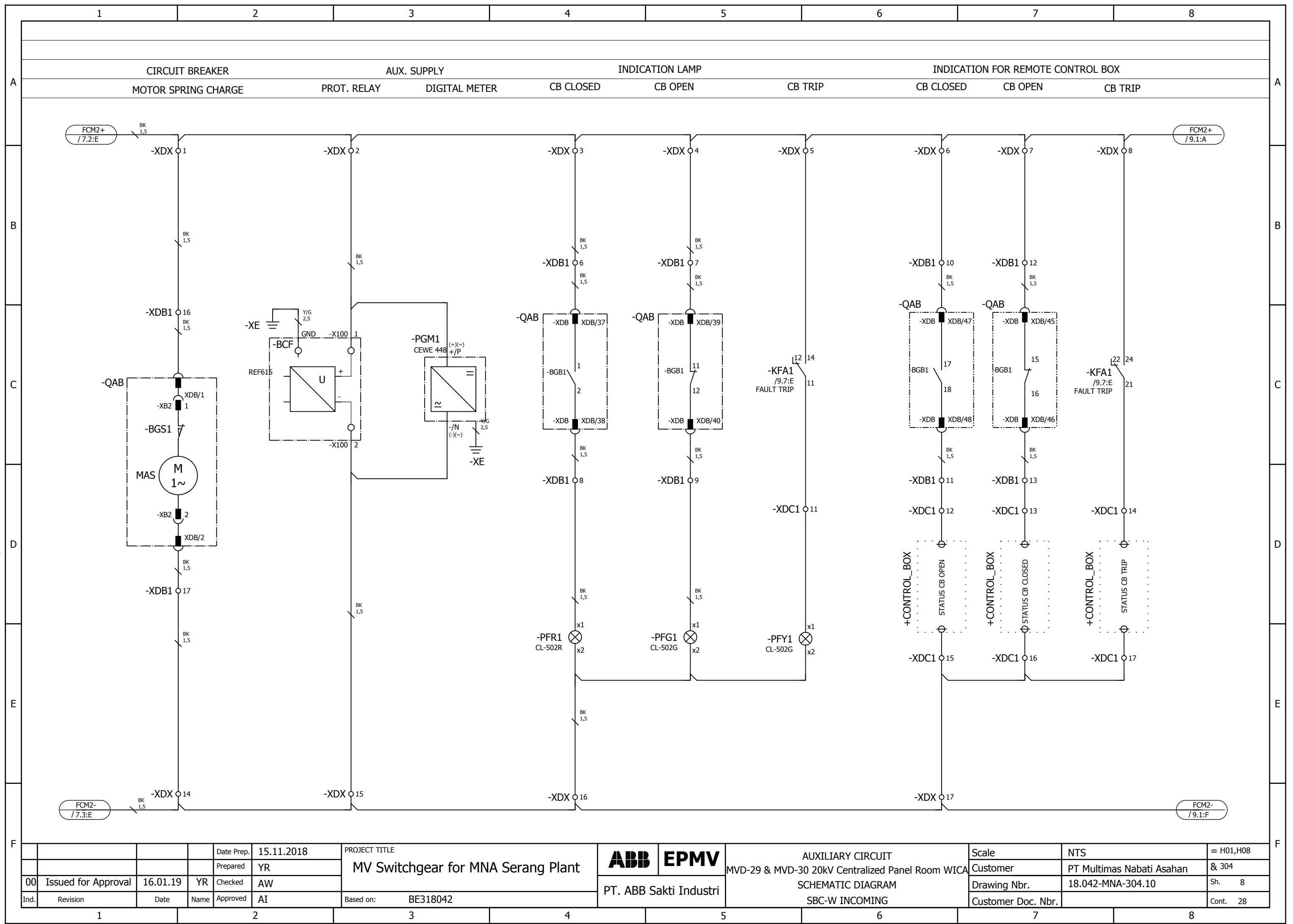
				Date Prep.	15.11.2018	PROJECT TITLE		 	REFERENCE DESIGNATIONS MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS	= H01,H08
				Prepared	YR	MV Switchgear for MNA Serang Plant				Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	16.01.19	YR	Checked	AW					Drawing Nbr.	18.042-MNA-304.10	Sh. 6
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		PT. ABB Sakti Industri	SBC-W INCOMING	Customer Doc. Nbr.		Cont. 28

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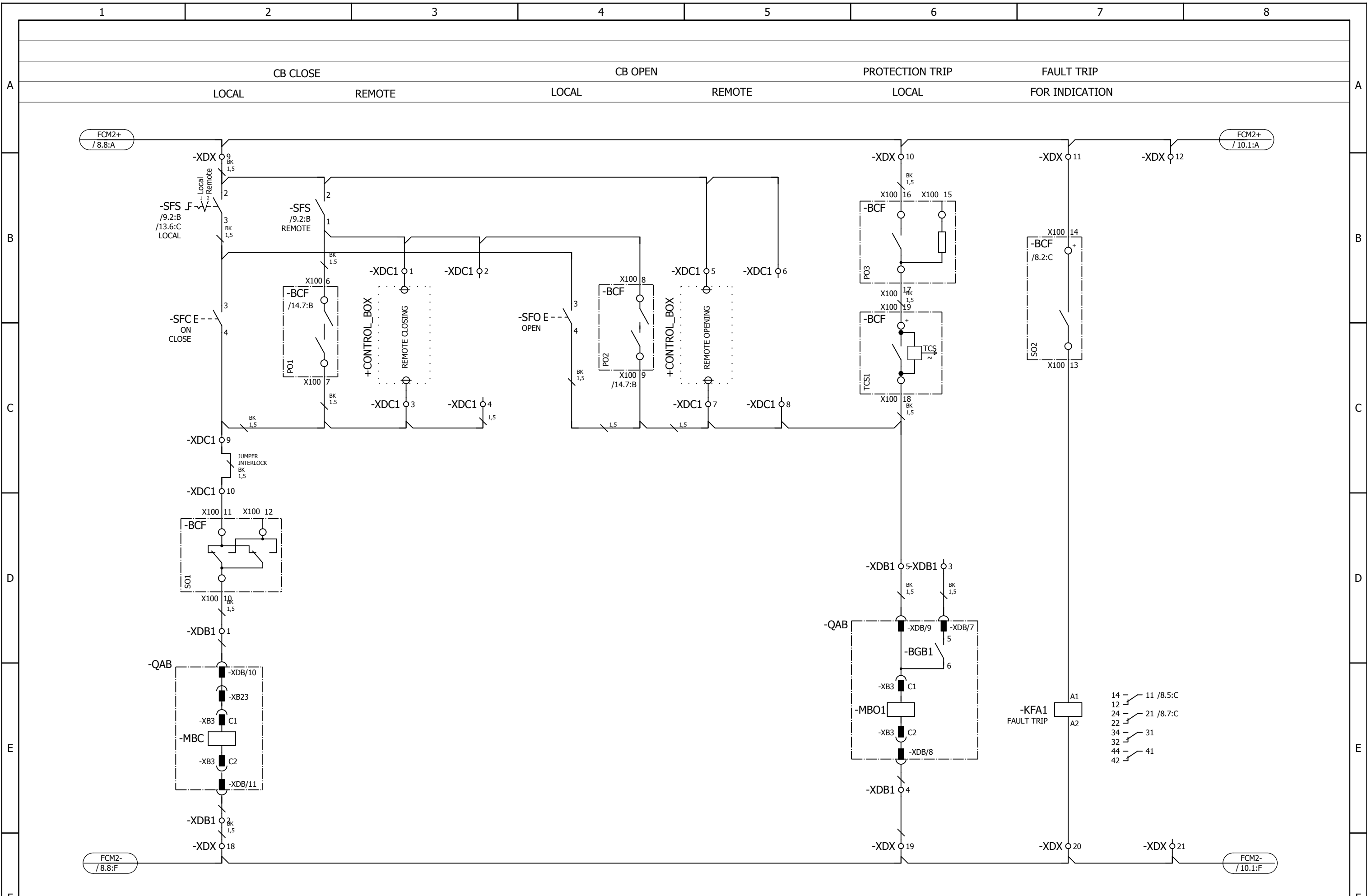
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	CIRCUIT BREAKER CONTROL MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS	= H01,H08
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	16.01.19	YR	Checked	AW					Drawing Nbr.	18.042-MNA-304.10	Sh. 9
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri		SCHEMATIC DIAGRAM SBC-W INCOMING	Customer Doc. Nbr.		Cont. 28

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	 	AUXILIARY MOTOR & RELAY MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS	= H01,H08
				Prepared	YR				Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	16.01.19	YR	Checked	AW		PT. ABB Sakti Industri	SCHEMATIC DIAGRAM SBC-W INCOMING	Drawing Nbr.	18.042-MNA-304.10	Sh. 10
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042			Customer Doc. Nbr.		Cont. 28

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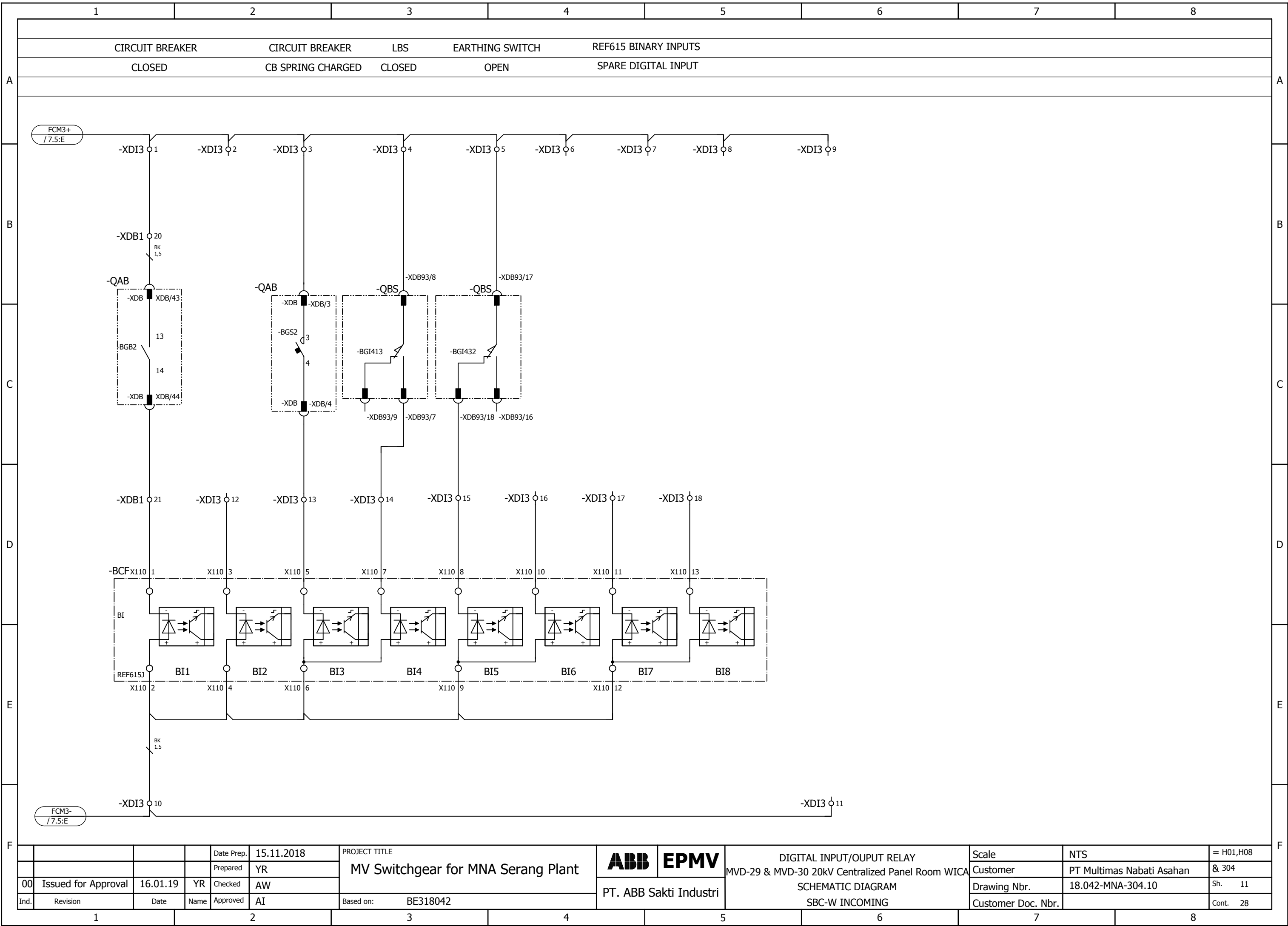


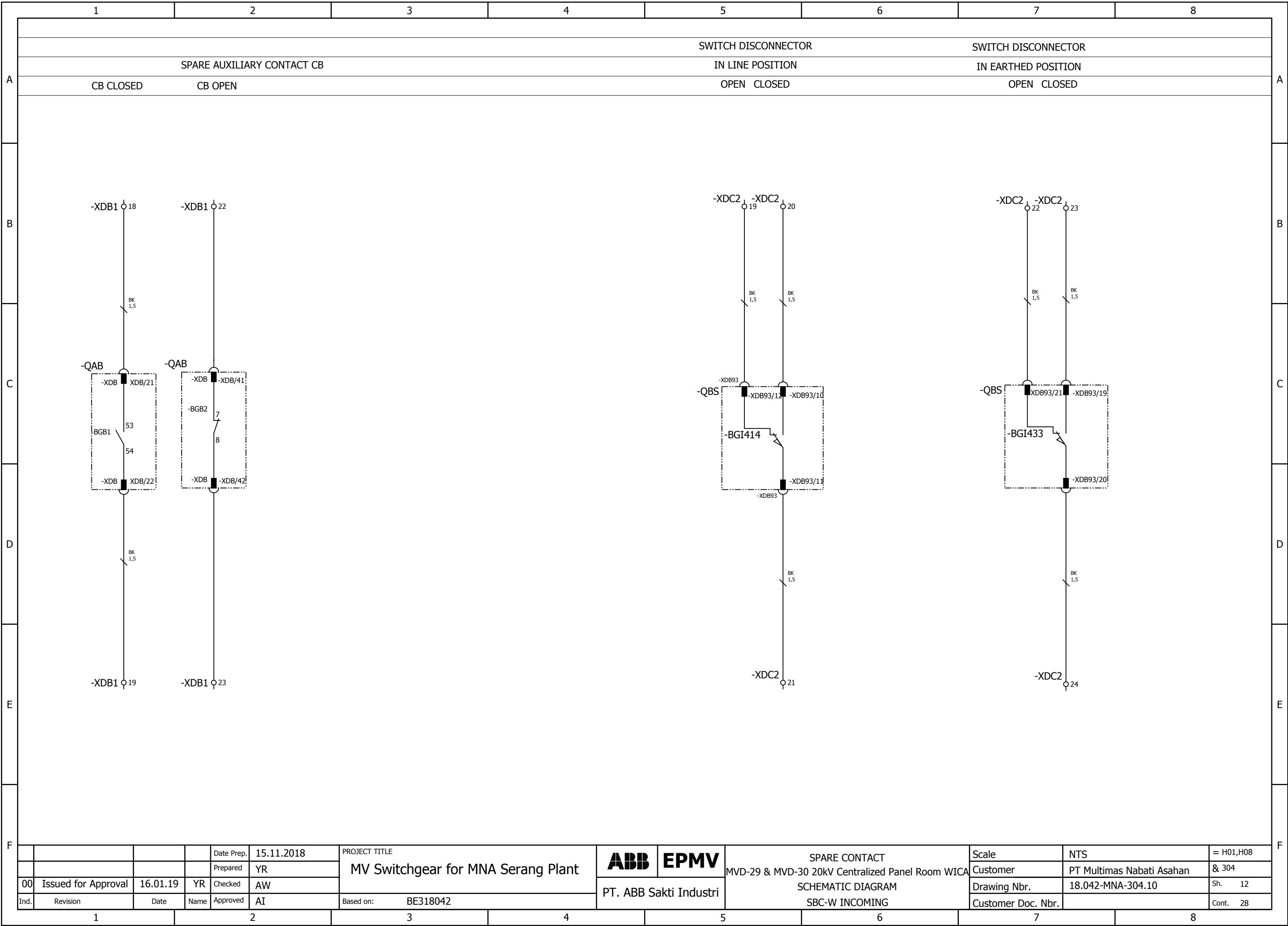
ABB **EPMV**

PT. ABB Sakti Industri

DIGITAL INPUT/OUTPUT RELAY
MVD-29 & MVD-30 20kV Centralized Panel Room WICA
SCHEMATIC DIAGRAM
SBC-W INCOMING

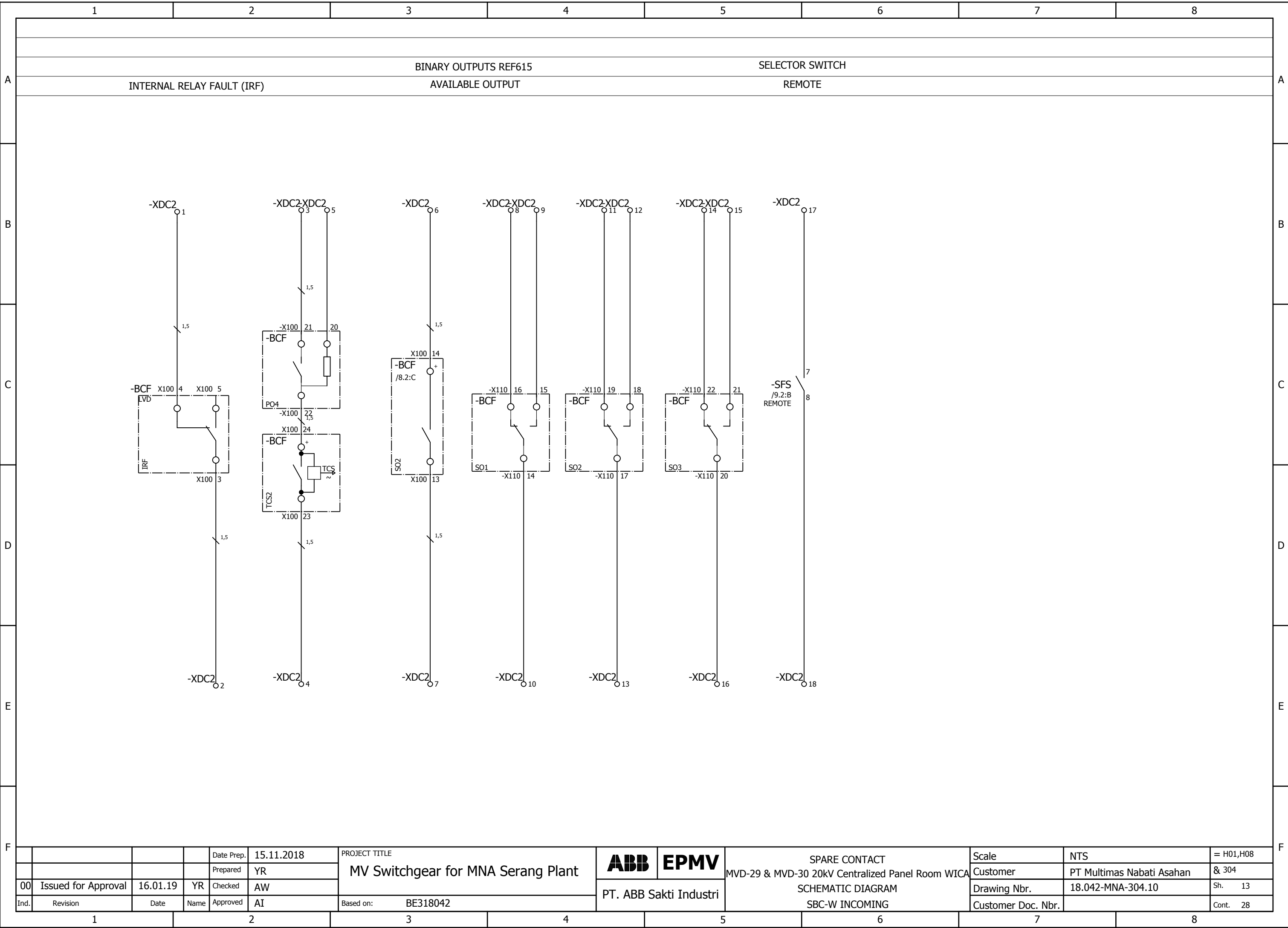
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Customer	PT Multimas Nabati Asahan	& 304
Drawing Nbr.	18.042-MNA-304.10	Sh. 11
Customer Doc. Nbr.		Cont. 28

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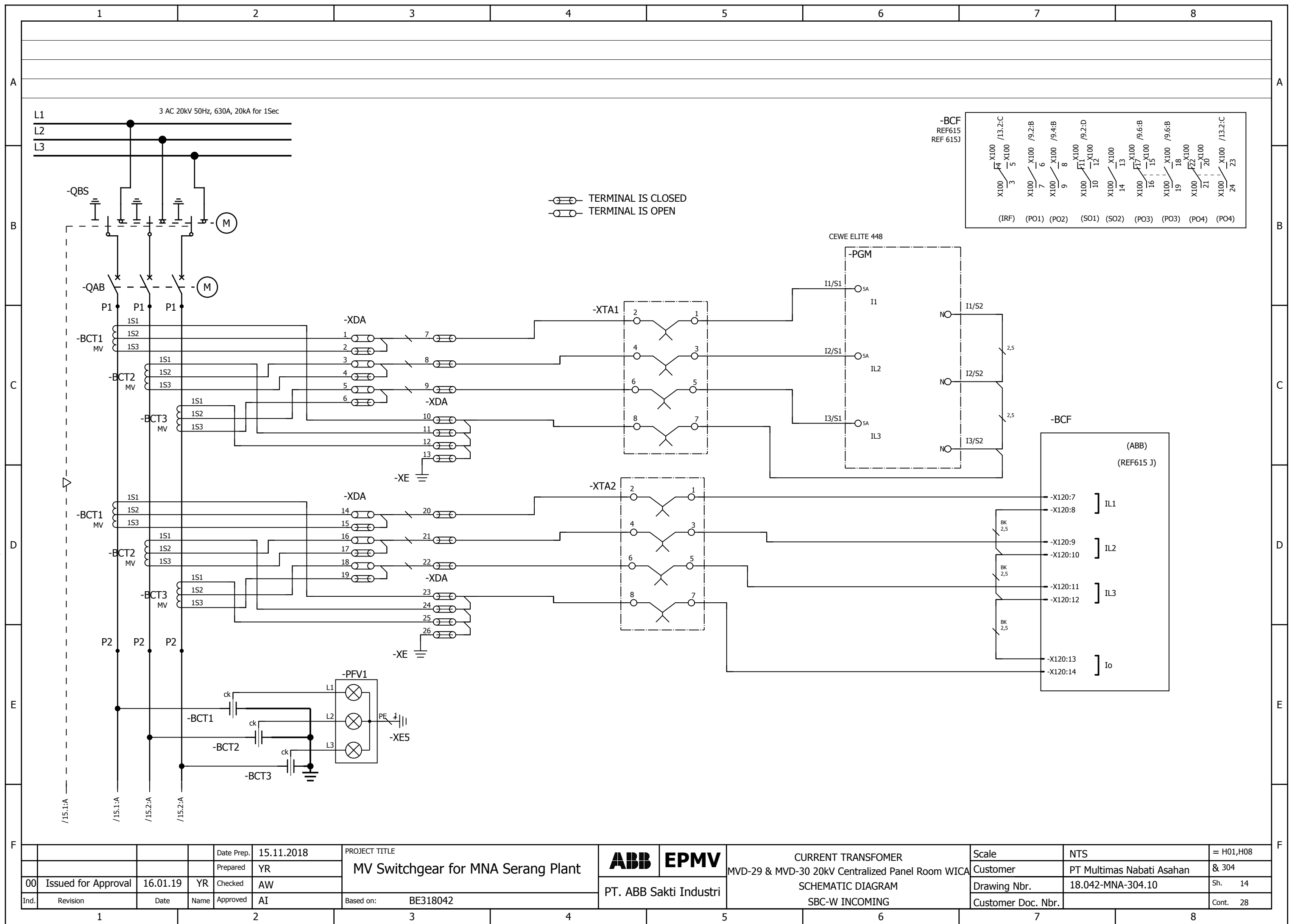
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SPARE CONTACT MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS	= H01,H08
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	16.01.19	YR	Checked	AW		PT. ABB Sakti Industri	SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-304.10	Sh.
Ind.	Revision	Date	Name	Approved	AI	Based on:		SBC-W INCOMING	Customer Doc. Nbr.		Cont.	28

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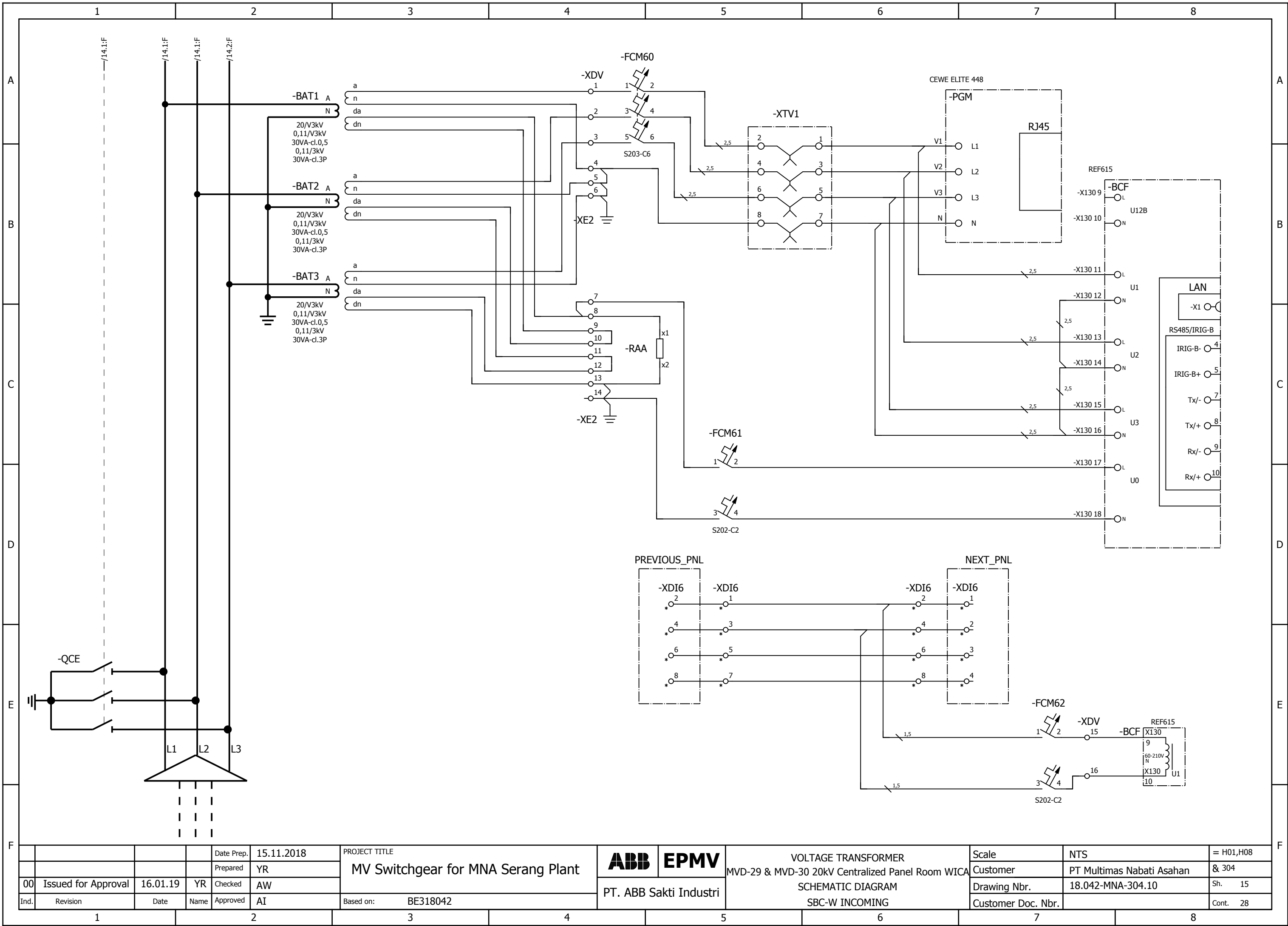


				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SPARE CONTACT MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS	= H01,H08
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	16.01.19	YR	Checked	AW		PT. ABB Sakti Industri	SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-304.10	Sh.
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		SBC-W INCOMING	Customer Doc. Nbr.		Cont.	28

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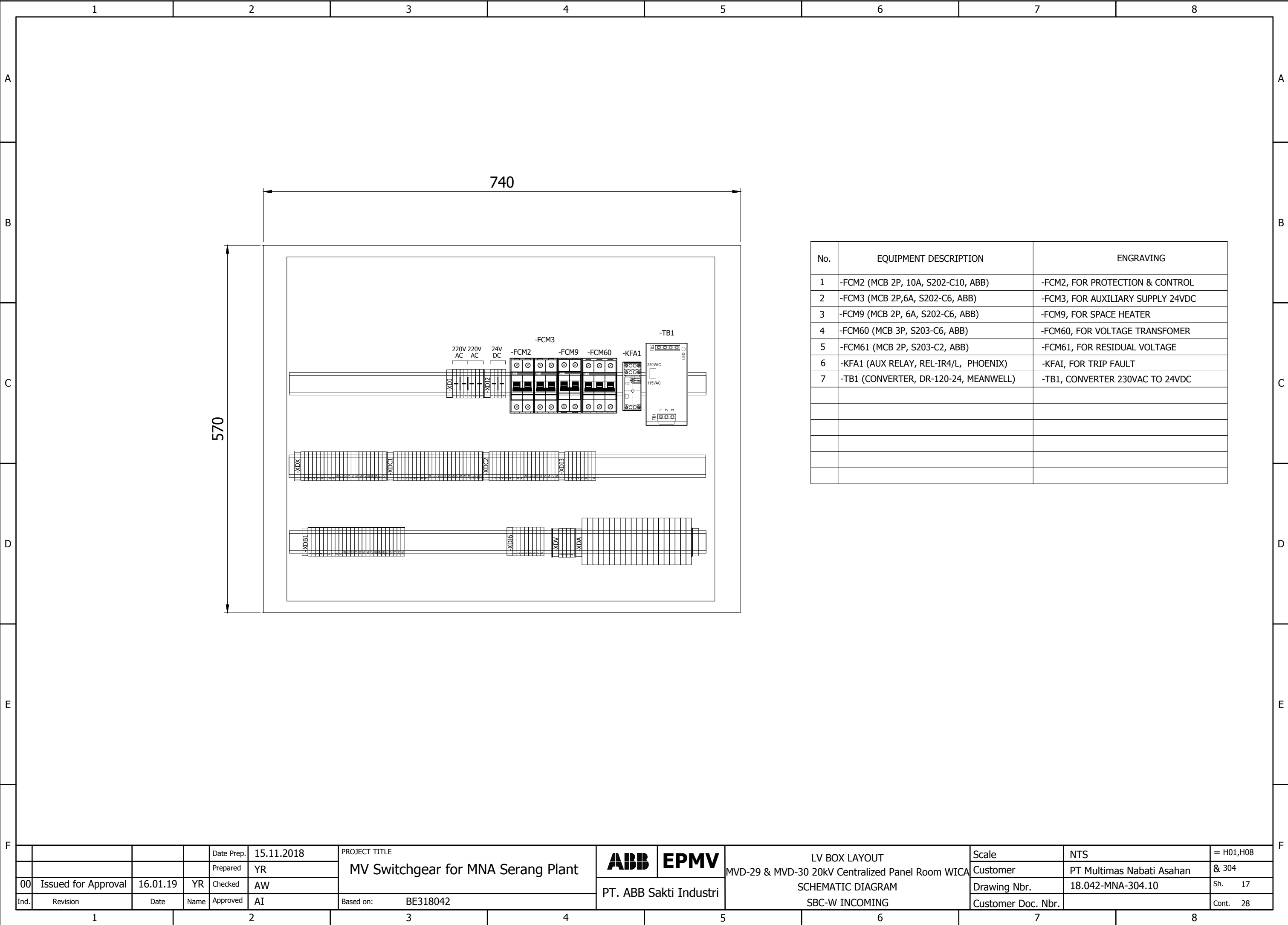


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No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK
10	-XTV1 (TEST BLOCK , YONGSUNG)	-XTV, VOLTAGE TEST BLOCK

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				Date Prep.	15.11.2018	PROJECT TITLE		ABB EPMV	LV BOX LAYOUT MVD-29 & MVD-30 20kV Centralized Panel Room WICA SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01,H08	
				Prepared	YR	MV Switchgear for MNA Serang Plant				Customer	PT Multimas Nabati Asahan	& 304	
00	Issued for Approval	16.01.19	YR	Checked	AW					Drawing Nbr.	18.042-MNA-304.10	Sh.	17
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont.	28

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Summarized parts list																IDABB_F02_005-IDABB	
Carry over: 0.00																	
TAG ID		DESCRIPTION				MANUFACTUR/BRAND		PART NUMBER		QUANTITY		REMARKS					
-BCF		Feeder Management Relay HBFJAEAGABC1ANA11G				ABB		REF615J HBFJAEAGABC1ANA11G		2		/8.2:C;/15.7:B					
-EB		HEATER 230VAC, 50W				SANYANG		CDXN010033P0009		1		/7.6:E					
-FCM2		MCB S202-C10 MCB. 2P, 10A				ABB		2CDS252001R0104		1		/7.2:C					
-FCM3;-FCM9		MCB S202-C6 MCB. 2P, 6A				ABB		2CDS252001R0064		2		/7.5:C;/7.6:C					
-FCM60		MCB S203-C6 MCB. 3P, 6A				ABB		2CDS253001R0064		1		/15.4:A					
-FCM61;-FCM62		MCB S202-C2 MCB. 2P, 2A				ABB		2CDS252001R0024		2		/15.5:D;/15.7:E					
-KFA1		Auxiliary Relay 230AC 4COS				PHOENIX		1YRA3DL266230001		1		/9.7:E					
-PFG1		PILOT LIGHT GREEN #CL-523G#230VAC CL-523G				ABB		1SFA619402R5232		1		/8.5:E					
-PFR1		PILOT LIGHT RED #CL-523R#230VAC CL-523R				ABB		1SFA619402R5231		1		/8.4:E					
-PFY1		PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y				ABB		1SFA619402R5233		1		/8.6:E					
-PGM1		POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC				CEWE		1YRA4EA0350001		1		/8.3:C					
-SFC		PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR				IDEC		1YRA3DA14700001		1		/9.2:B					
-SFO		PUSH BUTTON, YW1B-M1E10-R RED COLOUR				IDEC		1YRA3DA14700002		1		/9.4:B					
-SFS		LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)				K&N		1YRA3DA17940011		1		/9.2:B					
-TB1		Voltage Input : 85-264 VAC/120-370V DC Voltage Output : 24 VDC, 5A				MEANWELL		DR-120-24		1		/7.3:D					
-XTA1;-XTA2		Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSCTT-04C		2		/14.4:B;/14.4:D					
-XTV1		Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSPTT-04C		1		/15.5:A					

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Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT Multimas Nabati Asahan	= H01,H08 & 304 Sh. 25 Cont. 28
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB	EPMV	PT. ABB Sakti Industri	Terminal diagram	Scale	Customer	PT Multimas Nabati Asahan	= H01,H08 & 304 Sh. 25 Cont. 28
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